

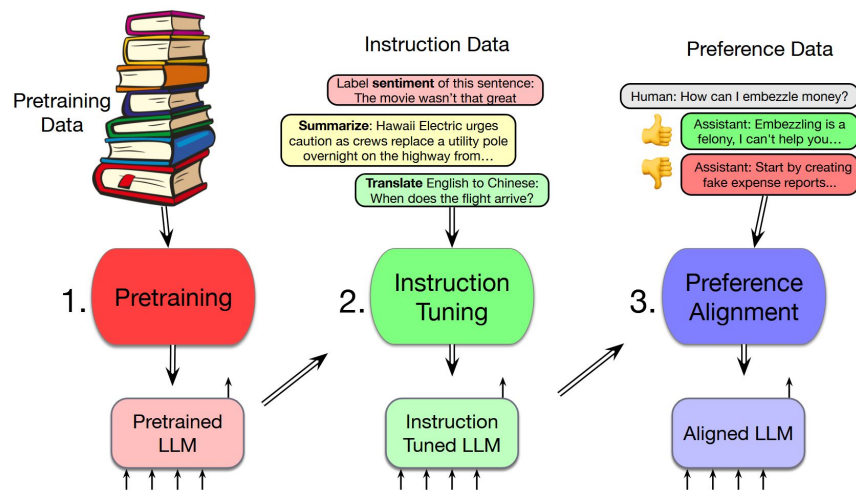
Large Language Models for the
Economic and Social Sciences:

Demystifying LLMs III

Indira Sen, HWS 2026

Last Time: How can we build LLMs?

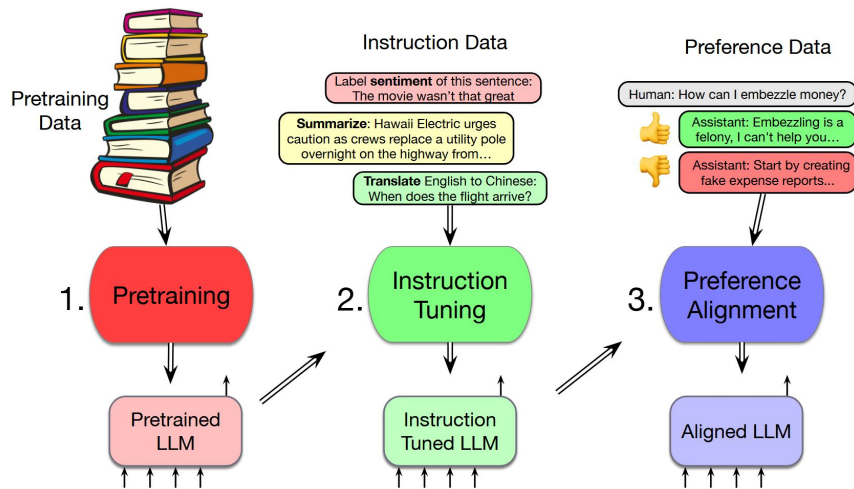
Somewhat simplified answer



Jurafsky and Martin, 2025

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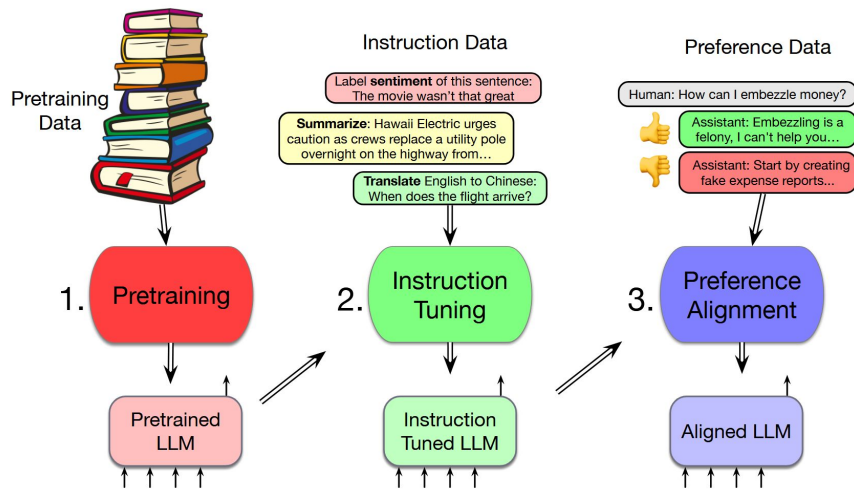
Jurafsky and Martin, 2025

Vastly Simplified answer

- We take a LOT of data encoding social, cultural, and of course linguistic traces
- Train NLP models that can generate content on this data
- We further train the model to follow and respond to instructions
- We further modify the model so that it doesn't have unwanted behavior

Last Time: How can we build LLMs?

Somewhat simplified answer



Jurafsky and Martin, 2025

Vastly Simplified answer

- We take a LOT of data encoding social, cultural, and of course linguistic traces
- **Train NLP models that can generate content on this data**
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Summary of Last Week and Today

Last session we learned:

- More efficient representations
—> word embeddings
- More powerful models for
classification —> deep learning
- Doing more with text
representations —> language
modeling
- Attention and Transformers

Today:

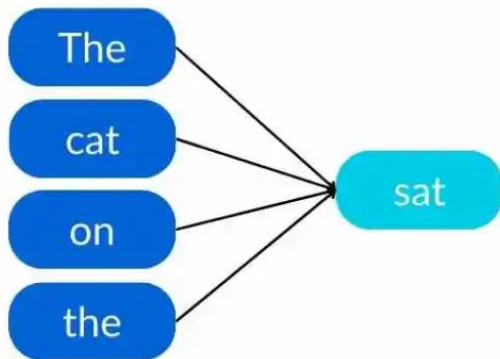
- Pretraining and fine-tuning
- Language generation
- Instruction-tuning
- Reinforcement Learning with
Human Feedback

Recap: Word embeddings

Example Sentence: The cat sat on the mat.

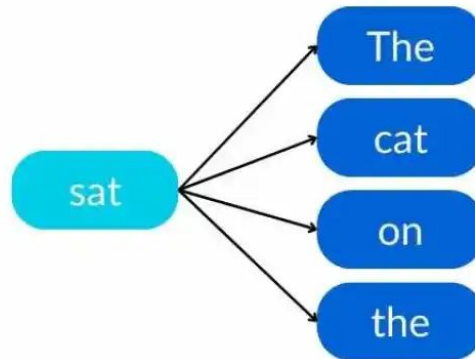
Continuous Bag-of-Words (CBOW)

Goal: Given context words,
predict the target word.



Skip-gram Model

Goal: Given a word,
predict the surrounding context words.



Attention

Attention

The fixed encoding

enco

Published as a conference paper at ICLR 2015

NEURAL MACHINE TRANSLATION BY JOINTLY LEARNING TO ALIGN AND TRANSLATE

Dzmitry Bahdanau

The idea behind 'Attention': At different steps, let a model "focus" on different parts of the input.

Я видел
"I" "saw"

tion. Unlike the traditional statistical machine translation, the neural machine translation aims at building a single neural network that can be jointly tuned to maximize the translation performance. The models proposed recently for neural machine translation often belong to a family of encoder-decoders and encode source sentence into a fixed-length vector from which a decoder generates a translation. In this paper, we conjecture that the use of a fixed-length vector is a bottleneck in improving the performance of this basic encoder-decoder architecture, and propose to extend this by allowing a model to automatically (soft-)search

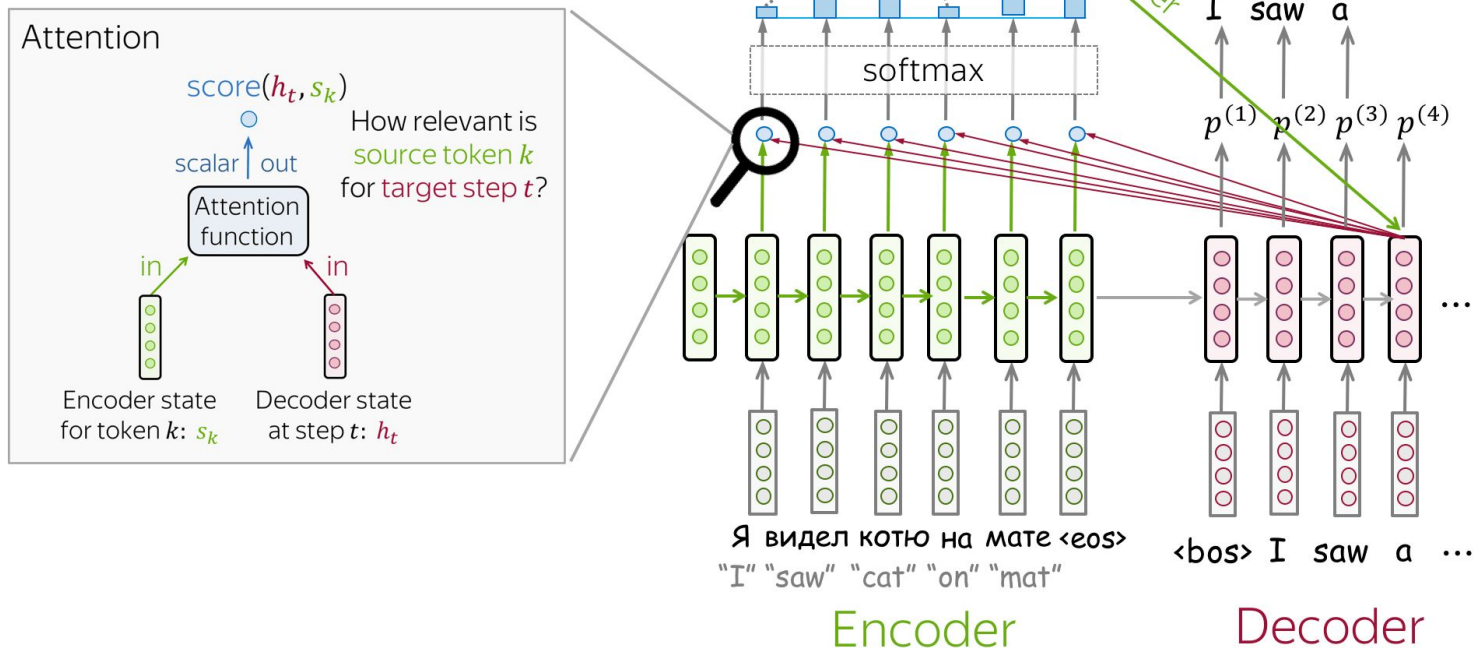
Bahdanau, Dzmitry, Kyunghyun Cho, and Yoshua Bengio. "[Neural machine translation by jointly learning to align and translate.](#)" *arXiv preprint arXiv:1409.0473* (2014).

Attention

Attention output: weighted sum of encoder states with attention weights

Attention weights: distribution over source tokens

A model can learn to “pay attention” to the most relevant source tokens for each step



Contextual Embeddings

Contextualized Embeddings (ELMo, CoVe)

Deep contextualized word representations

Matthew E. Peters[†], Mark Neumann[†], Mohit Iyyer[†], Matt Gardner[†],
{matthewp, markn, mohiti, mattg}@allenai.org

Christopher Clark*, Kenton Lee*, Luke Zettlemoyer^{†*}

The idea behind Contextualized
Embeddings: Word *token* vectors instead of
word *type*, with neural networks modeling
left and right contexts

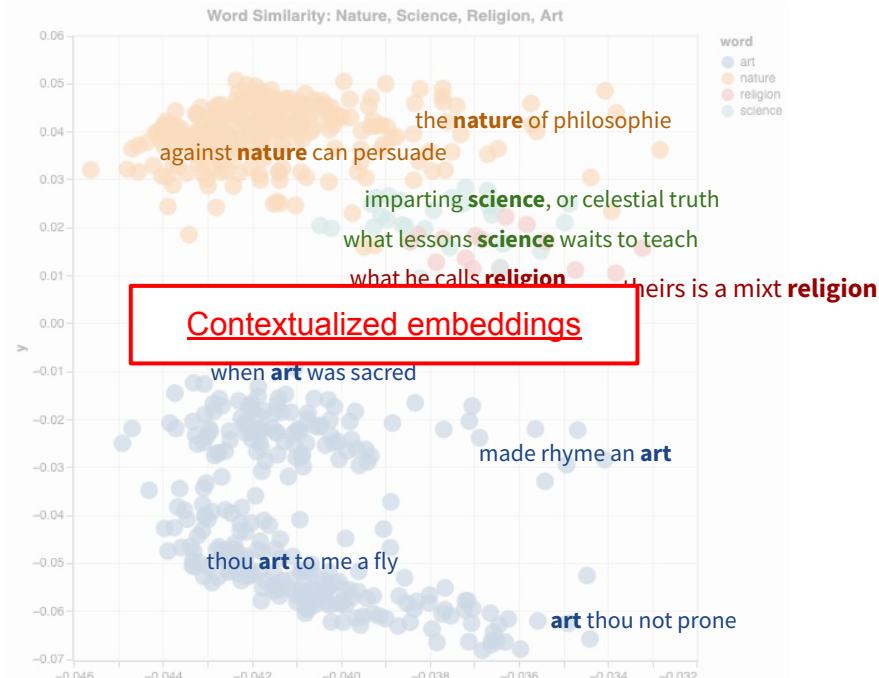
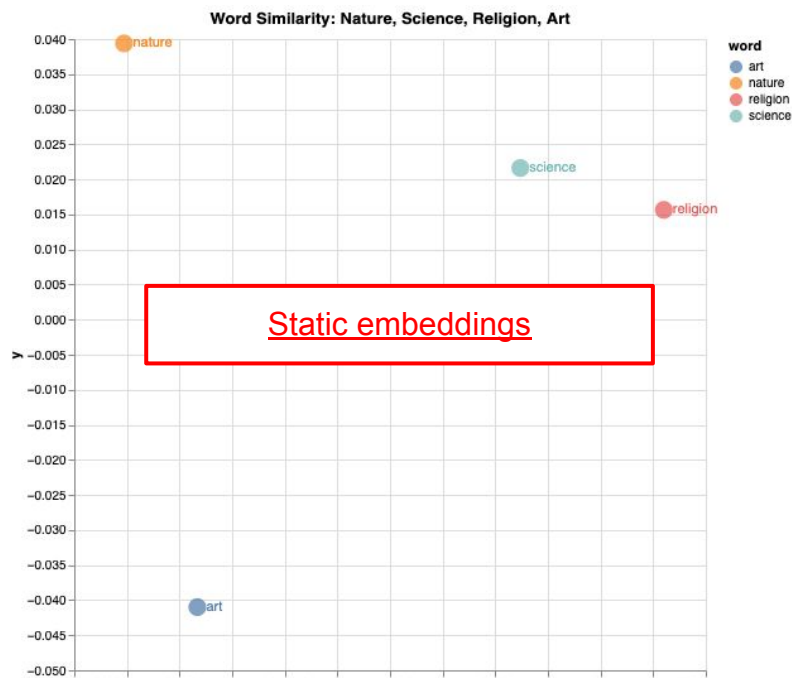
the 2018 Conference of the North American Chapter of the
Association for Computational Linguistics: Human Language
Technologies

contexts (i.e., to model
embeddings). Embeddings are learned func-
tions of the internal states of a deep bidirec-
tional language model (biLM), which is pre-
trained on a large text corpus. We show that
these representations can be easily added to
existing models and significantly improve the
state of the art across six challenging NLP

et al., 2017). ELMo representations are deep, in
the sense that they are a function of all of the in-
ternal layers of the biLM. More specifically, we
learn a linear combination of the vectors stacked
above each input word for each end task, which
markedly improves performance over just using
the top LSTM layer.

Contextualized Embeddings (ELMo, CoVe)

Result: multiple vectors of a word depending on its context



Transformers

Transformers

Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

The idea behind ‘Transformers’: Replace all components with Attention to model context better

.com

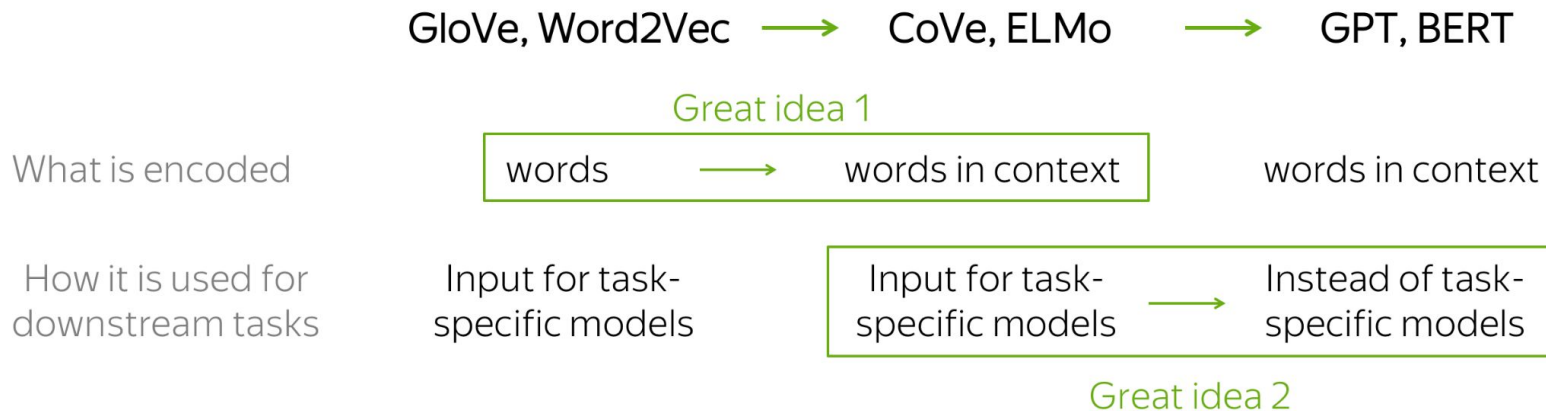
Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including

Vaswani, Ashish, et al. "[Attention is all you need.](#)" *Advances in neural information processing systems* 30 (2017).

One model to rule them all: Pretraining and Fine-tuning

BERT, GPT (the first one),



BERT, GPT (the first one),

G

What is encoded

How it is used for
downstream tasks

Devlin, Jacob, et al. "[Bert: Pre-training of deep bidirectional transformers for language understanding.](#)" *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*. 2019.

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Jacob Devlin Ming-Wei Chang Kenton Lee Kristina Toutanova

Google AI Language

{jacobdevlin, mingweichang, kentonl, kristout}@google.com

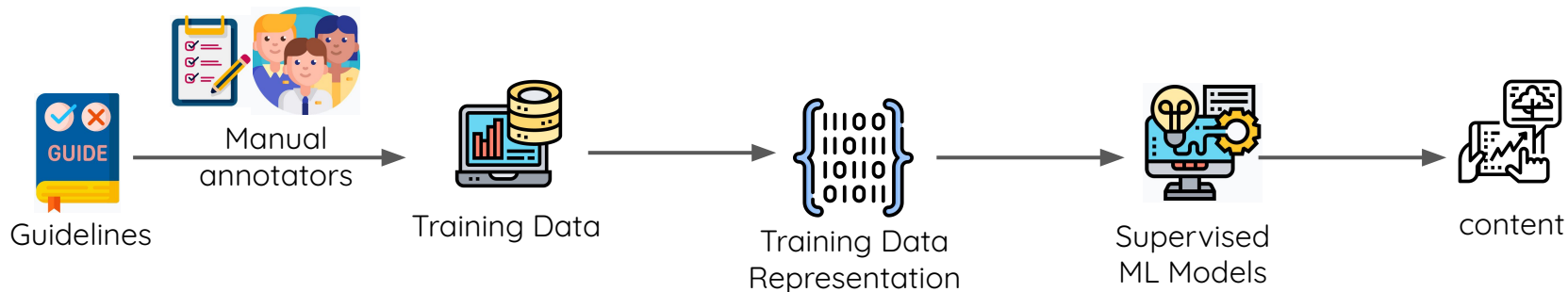
Abstract

We introduce a new language representation model called **BERT**, which stands for **B**idirectional **E**ncoder **R**epresentations from **T**ransformers. Unlike recent language representation models (Peters et al., 2018a; Radford et al., 2018), BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide

There are two existing strategies for applying pre-trained language representations to downstream tasks: *feature-based* and *fine-tuning*. The feature-based approach, such as ELMo (Peters et al., 2018a), uses task-specific architectures that include the pre-trained representations as additional features. The fine-tuning approach, such as the Generative Pre-trained Transformer (OpenAI GPT) (Radford et al., 2018), introduces minimal task-specific parameters, and is trained on the downstream tasks by simply fine-tuning *all* pre-trained parameters. The two approaches share the

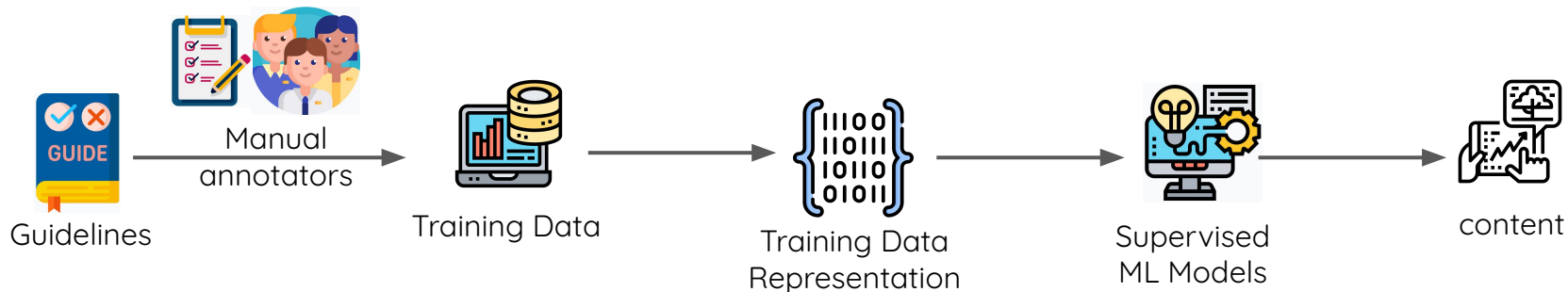
Revisiting Machine Learning

- Usual workflow, train model on your dataset for a certain task.
- The dataset has to represent exactly what task you want to do

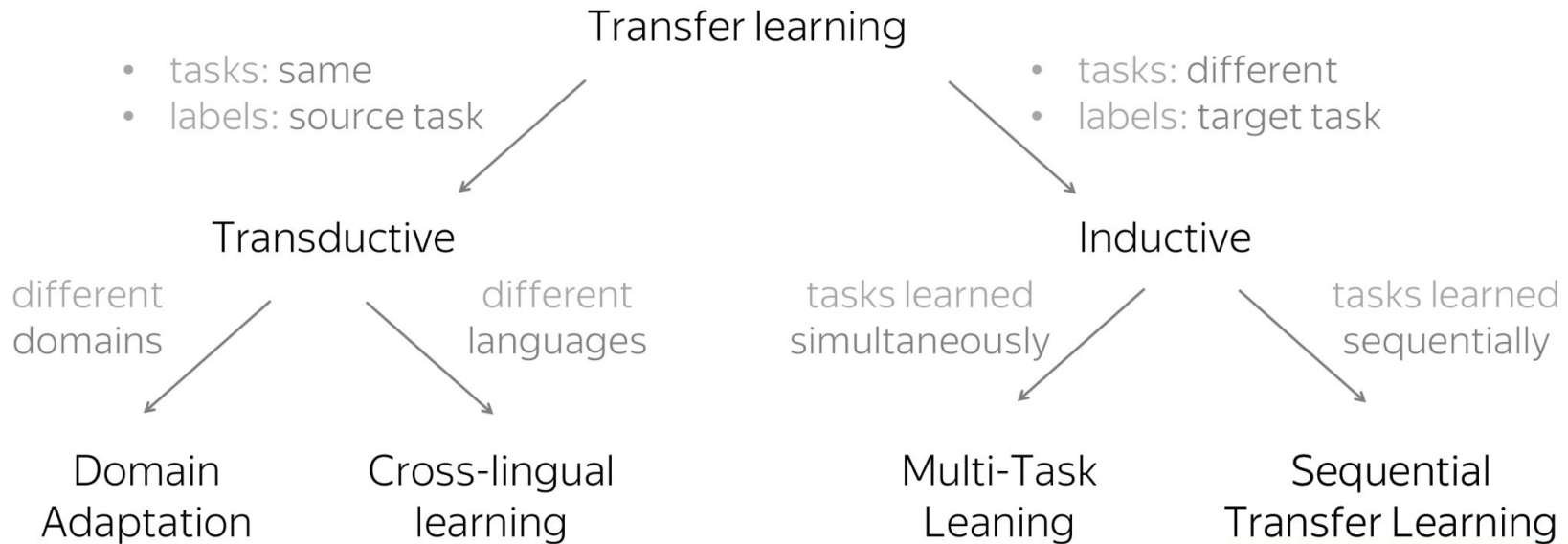


Revisiting Machine Learning

- Usual workflow, train model on your dataset for a certain task.
- The dataset has to represent exactly what task you want to do
- However,
 - Sometimes, you do not have enough labeled data for a very similar task
- Transfer Learning!



Revisiting Machine Learning: Transfer Learning



This taxonomy is from [Sebastian Ruder's blog post](#).

The old vs. 'new' paradigm

OLD: Train a model on a task where we have labeled data

NEW:

1. **Pre-train** a model on a task where we have lots of data
 - *e.g., we have lots and lots of internet data*
2. **Fine-tune** the model on your downstream task
 - *e.g., a specific, curated dataset that you're studying*

 **Transfer learning!**

Transfer Learning via Pretraining

- Train from scratch

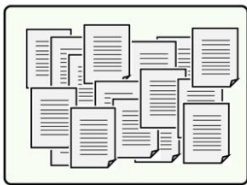
What they will know:



May be not enough to
learn relationships
between words

- Take pretrained
(Word2Vec, GloVe)

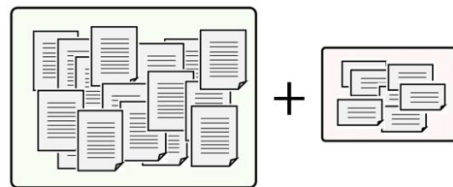
What they will know:



Know relationships between words,
but are **not** specific to the task

- Initialize with pretrained,
then fine-tune

What they will know:



Know relationships between
words and adapted for the task

“**Transfer**” knowledge from a huge unlabeled corpus to your task-specific model

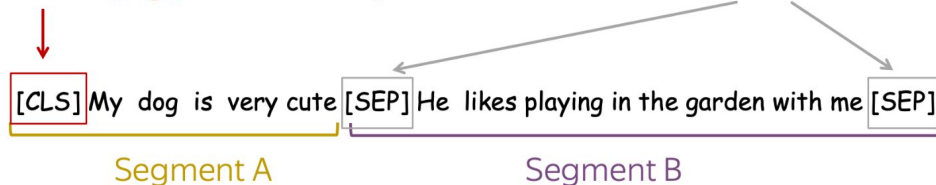
How is BERT Pre-trained?

- Pre-training Input: Pairs of Sentences with Special Tokens separating them

[CLS]: Special token

- Training time: predict if sentences are consecutive or not (Next Sentence Prediction /NSP objective)
- Test time: downstream tasks (e.g., classification)

[SEP]: Special token-separator



Training on pairs of sentences: either consecutive or random (50%/50%)

- Pre-training Objective:

- Next Sentence Prediction (NSP)
- Masked Language Modeling (MLM)

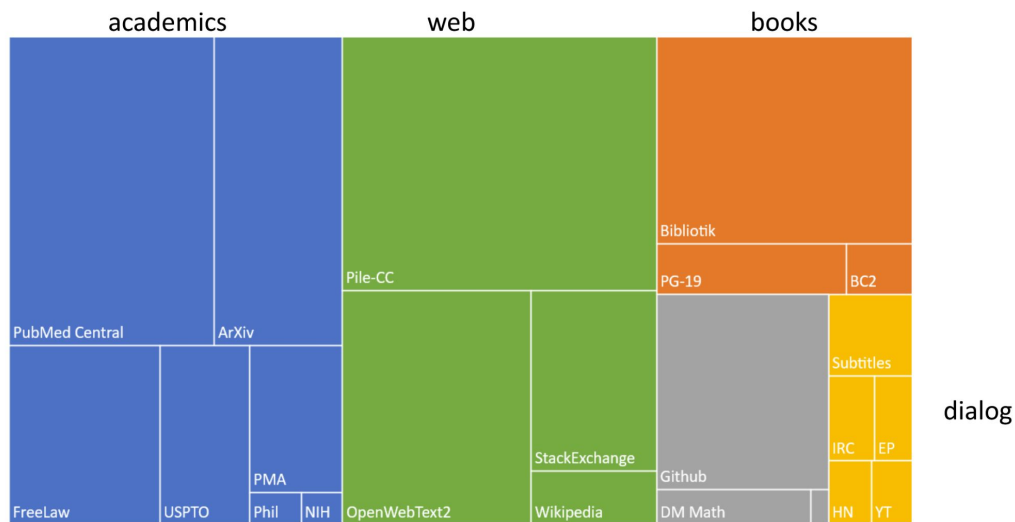
- Lena Voita, NLP Course for you

Massive and Undocumented Pretraining Sets

What kind of datasets are used to pre-train models like BERT?

- “BooksCorpus”
- English Wikipedia
- “The Pile”
- “Colossal Clean Crawled

The Pile: a pretraining corpus



How does fine-tuning work?

- Inputs:
 - pretrained model
 - Labeled dataset (rule-of-thumb, at least 1K examples)
- Adds a new layer containing classification parameters.
- All model parameters are updated to maximize the log probability of the labels.
- Output: classifier for your specific task



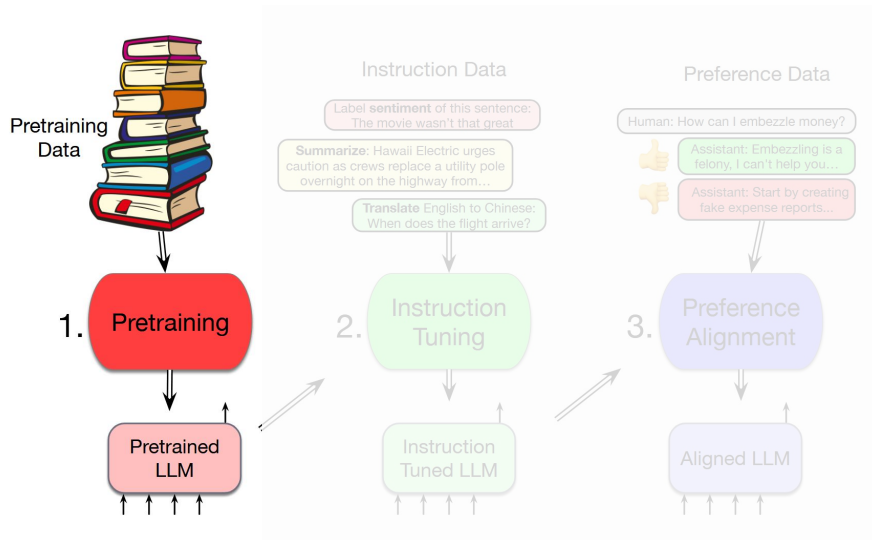
Image adapted from :
<https://aiforhumanists.com/workshops/>

Let's explore what BERT has learned from
pretraining

[https://pair.withgoogle.com/explorables/fil
l-in-the-blank/](https://pair.withgoogle.com/explorables/fil
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We now have (almost) all the ingredients for step 1

- Efficient representations:
contextualized embeddings
- Neural language model
- **Pretrained** in an unsupervised way
on large amounts of data

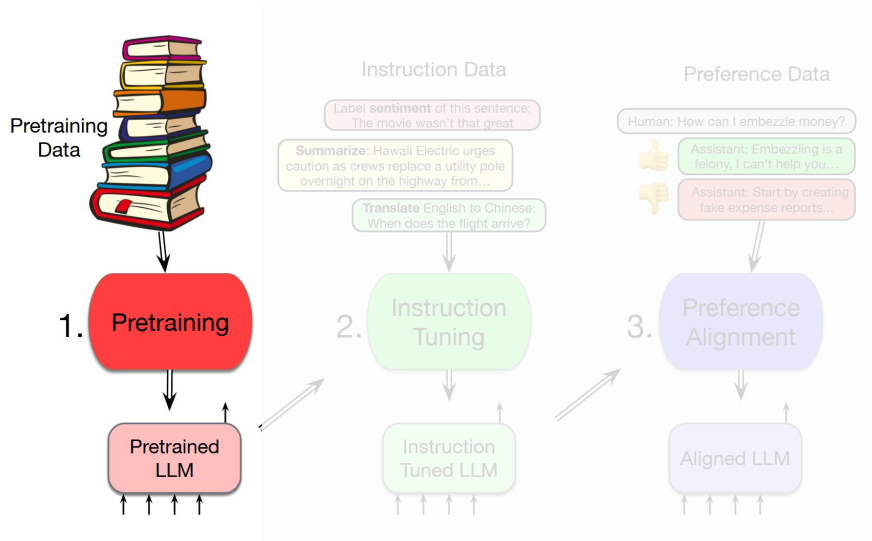


Jurafsky and Martin, 2025

Train NLP models that can generate content
after pretraining on large datasets

We now have (almost) all the ingredients for step 1

- Efficient representations:
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- Neural language model
- **Pretrained** in an unsupervised way on large amounts of data
- BERT-style models only have encoders → cannot generate content



Jurafsky and Martin, 2025

Train NLP models that can generate content
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Language Modeling: Probability + Language

$P(\text{I saw a cat on } \dots) =$

$$\underbrace{P(\text{I}) \cdot P(\text{saw}|\text{I}) \cdot P(\text{a}|\text{I saw}) \cdot P(\text{cat}|\text{I saw a}) \cdot P(\text{on}|\text{I saw a cat})}_{\text{Probability of I saw a cat on}} \cdot \underbrace{\dots}_{\text{...}}$$



Probability of I saw a cat on

Formally, let $y_1, y_2, \dots, y_n$ be tokens in a sentence, and $P(y_1, y_2, \dots, y_n)$ the probability to see all these tokens (in this order). Using the product rule of probability (aka the chain rule), we get

$$P(y_1, y_2, \dots, y_n) = P(y_1) \cdot P(y_2|y_1) \cdot P(y_3|y_1, y_2) \cdot \dots \cdot P(y_n|y_1, \dots, y_{n-1}) = \prod_{t=1}^n P(y_t|y_{<t}).$$

Natural Language Generation (NLG)

NLG focuses on systems that produce fluent, coherent and useful language output for human consumption

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Non-open-ended generation: the input mostly determines the output generation.

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NLG focuses on systems that produce fluent, coherent and useful language output for human consumption

Open-ended generation: the output distribution still has high freedom

Non-open-ended generation: the input mostly determines the output generation.



How does an LLM generate content?

- Models like BERT are **encoder** only models
 - Are usually finetuned (trained on supervised data) for classification tasks.
- For generative tasks, we use either **decoder** models like GPT, LLaMa
 - It takes as input a series of tokens, and iteratively generates an output token one at a time.
 - Left to right (causal, **autoregressive**)
- Another family of generative models: **encoder-decoder** models like Flan T5, Whisper
 - Trained to map from one sequence to another
 - Useful for machine translation, speech recognition

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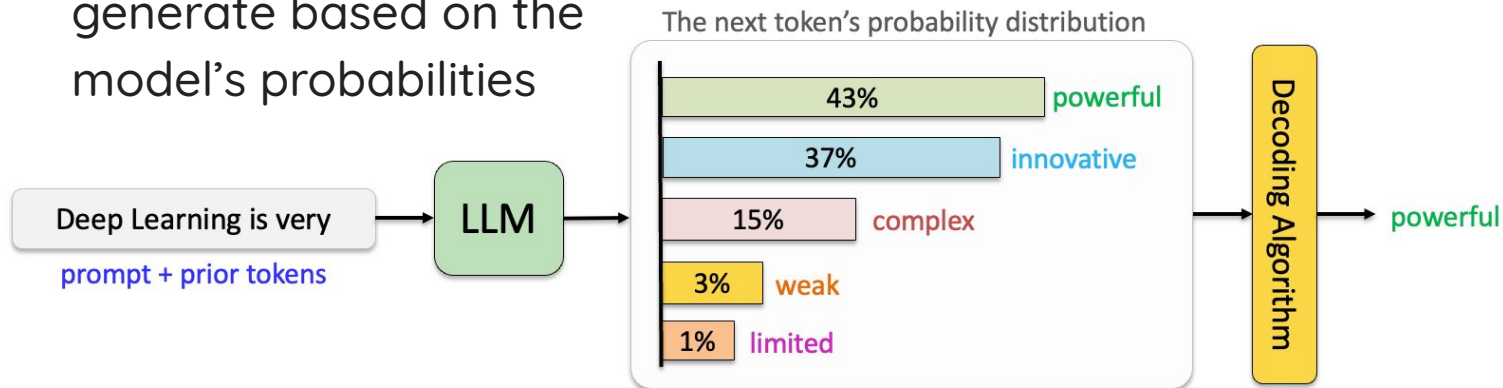
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How does an LLM generate content?

- Conditioned on previous contexts, e.g., some type of input or **prompt**
 - computing the probability of the next token from the prior context
 - sampling from that distribution to generate a token

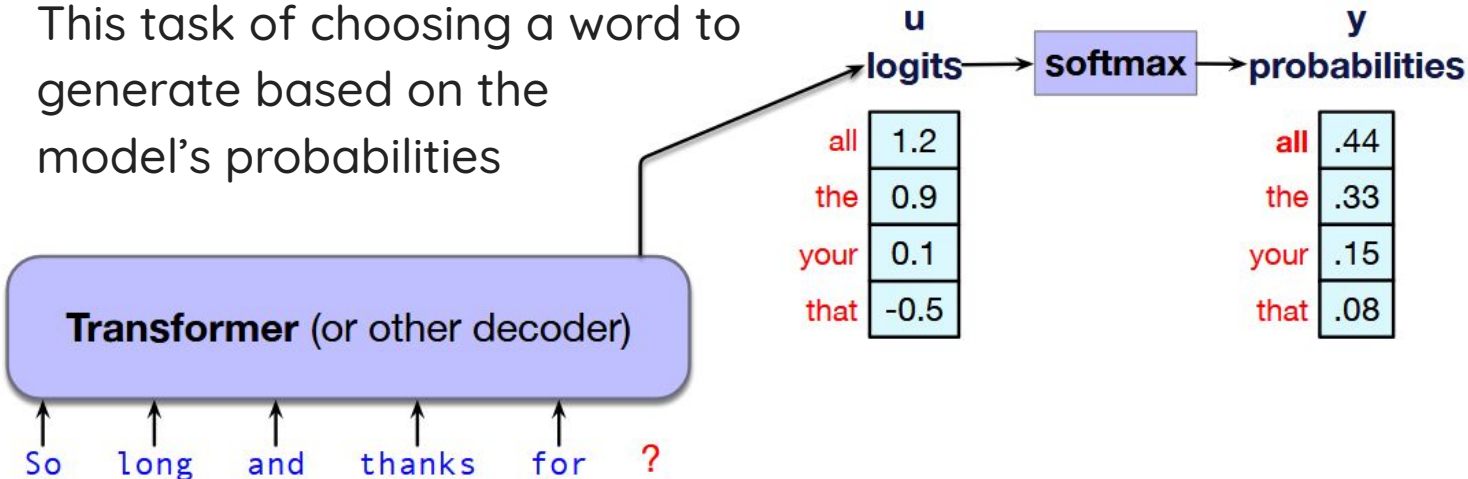
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- Decoding
 - This task of choosing a word to generate based on the model's probabilities



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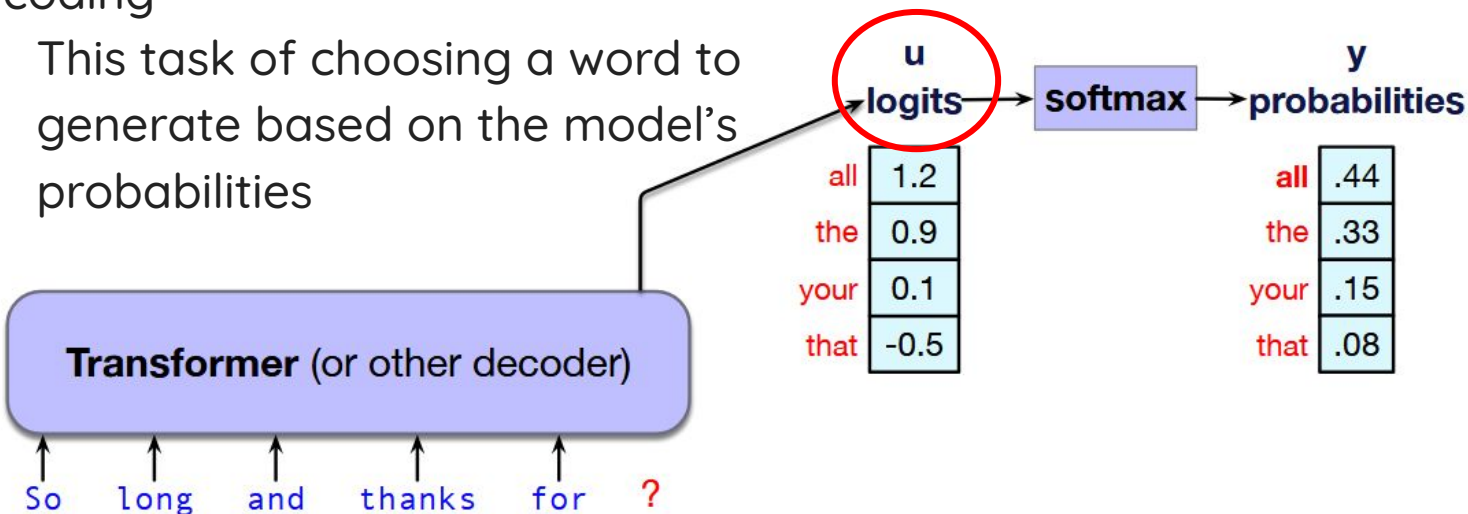


How does an LLM generate content?

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The internal networks for LLMs generate real-valued scores called logits for each token in the vocabulary.

- Decoding
 - This task of choosing a word to generate based on the model's probabilities

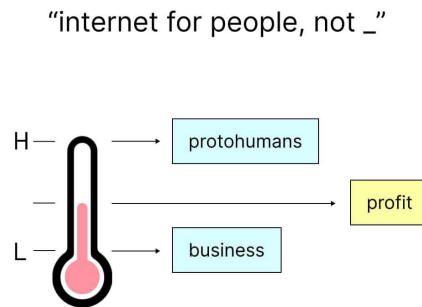


Decoding Algorithms (more details in lecture 4)

- Greedy decoding: pick the most high probability token
- Random sampling: pick any token at random
- Temperature sampling: increase the probability of picking higher probability tokens

Decoding Algorithms (more details in lecture 4)

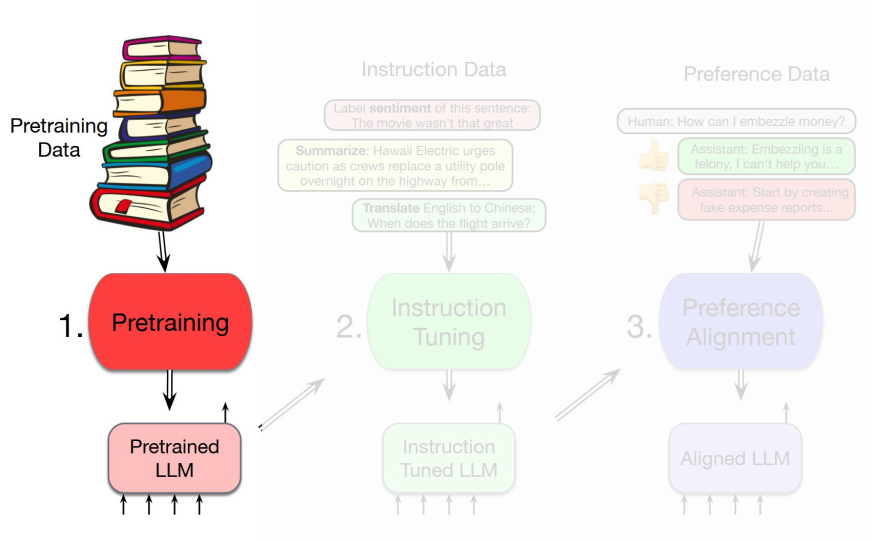
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Have you tried changing the temperature when interacting with LLMs?

We now have all the ingredients for step 1

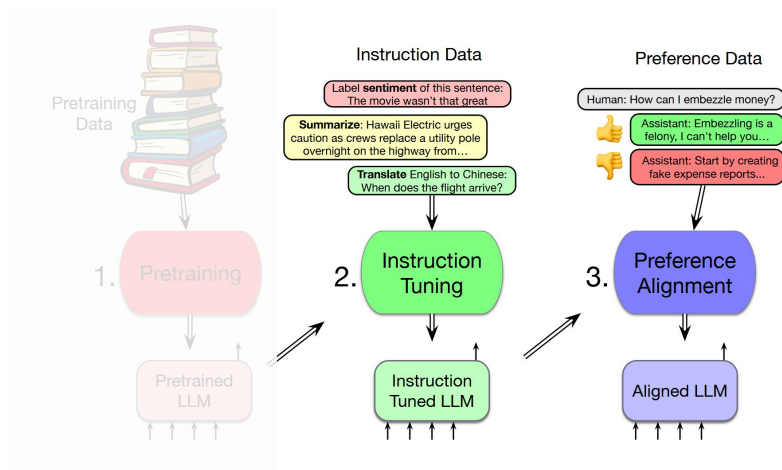
- Efficient representations:
contextualized embeddings
- Neural language model
- **Pretrained** in an unsupervised way on large amounts of data
- Models that generate context conditioned on previous tokens



Jurafsky and Martin, 2025

Train NLP models that can generate content
after pretraining on large datasets

Let's try some text generation ourselves:
[https://banana-projects-transformer-auto
complete.hf.space/doc/gpt2-large](https://banana-projects-transformer-auto-complete.hf.space/doc/gpt2-large)



After pretraining, comes **posttraining**, i.e.

1. Instruction tuning (also called supervised finetuning or SFT)
2. Preference alignment

Instruction Finetuning

Before: sentence completion, not answering a prompt.

Language models are not aligned with user intent

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

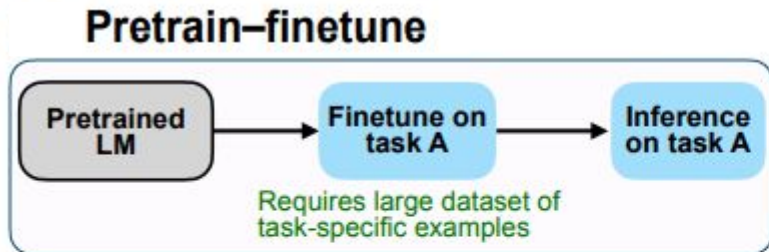
Explain evolution to a 6 year old.

Instruction Finetuning

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Instruction Finetuning!



[Finetuned Language Models Are Zero-Shot Learners](#)

Training language models to follow instructions with human feedback

Long Ouyang* Jeff Wu* Xu Jiang* Diogo Almeida* Carroll L. Wainwright*

Pamela Mishkin* Chong Zhang Sandhini Agarwal Katarina Slama Alex Ray

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Ryan Lowe*

OpenAI

Abstract

Making language models bigger does not inherently make them better at following a user's intent. For example, large language models can generate outputs that are untruthful, toxic, or simply not helpful to the user. In other words, these models are not *aligned* with their users. In this paper, we show an avenue for aligning language models with user intent on a wide range of tasks by fine-tuning with human feedback. Starting with a set of labeler-written prompts and prompts submitted through a language model API, we collect a dataset of labeler demonstrations of the desired model behavior, which we use to fine-tune GPT-3 using supervised learning. We then collect a dataset of rankings of model outputs, which we use to further fine-tune this supervised model using reinforcement learning from human feedback. We call the resulting models *InstructGPT*. In human evaluations on our prompt distribution, outputs from the 1.3B parameter InstructGPT model are preferred to outputs from the 175B GPT-3, despite having 100x fewer parameters. Moreover, InstructGPT models show improvements in truthfulness and reductions in toxic content generation while having minimal performance regression on public

[Training language models to follow instructions with human feedback](#) show that fine-tuning with human feedback is a promising direction for aligning language models with human intent.

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Finetuned Language Models Are Zero-Shot Learners

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Training language models to follow instructions with human feedback

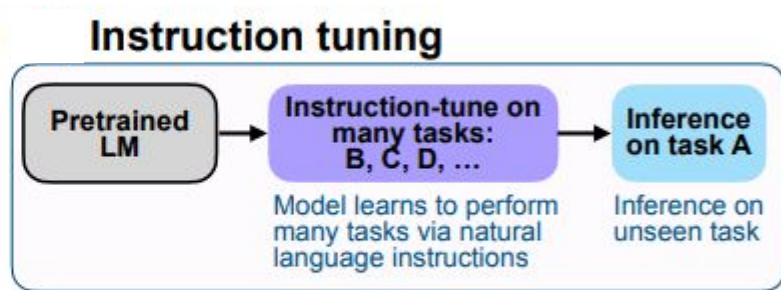
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Training language models to follow instructions with human feedback

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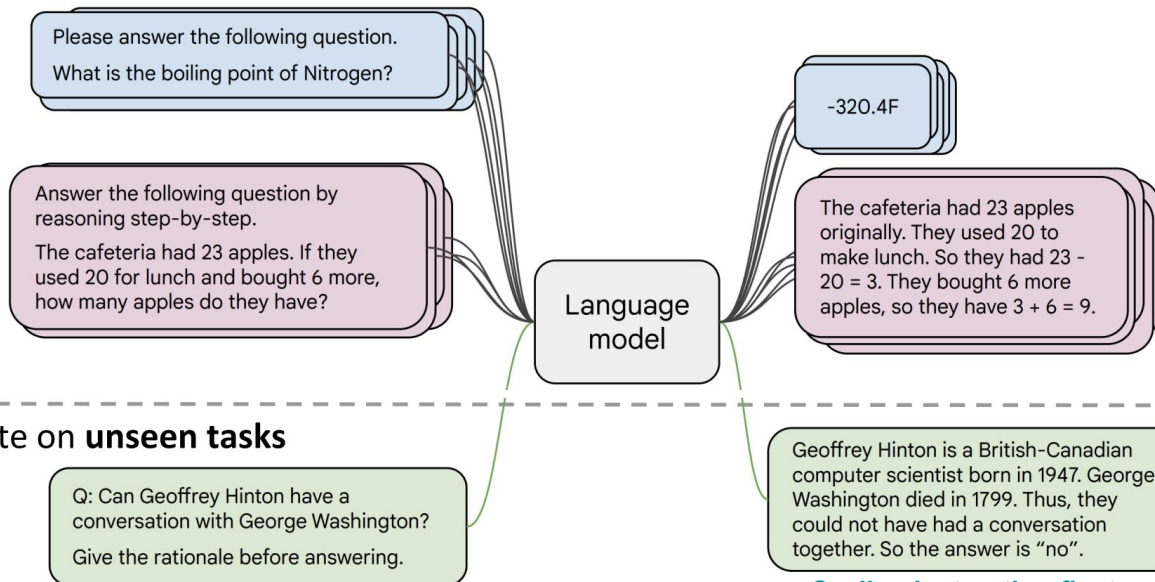
Instruction Finetuning

Also in Flan-T5

Scaling Instruction-Finetuned Language Models

Hyung Won Chung* Le Hou* Shayne Longpre* Barret Zoph† Yi Tay†
William Fedus† Yunxuan Li Xuezhi Wang Mostafa Dehghani Siddhartha Brahma
Albert Webson Shixiang Shane Gu Zhuyun Dai Mirac Suzgun Xinyun Chen
Aakanksha Chowdhery Alex Castro-Ros Marie Pellat Kevin Robinson
Dasha Valter Sharan Narang Gaurav Mishra Adams Yu Vincent Zhao
Yanning Huang Andrew Dai Hongkun Yu Slav Petrov Ed H. Chi
Jenny Zhou Quoc V. Le

- Collect examples of (instruction, output) pairs across many tasks and finetune an LM



Instructions have been shown to improve performance as we explore instruction finetuning across different model sizes, and (3) finetuning on these instructions dramatically improves performance on various prompting setups (zero-shot, few-shot, CoT), text-to-text generation, RealToxicityPrompts). The model performs PaLM 540B by a large margin on several benchmarks, such as MMLU, which achieve strong few-shot performance. Overall, instruction finetuning is a promising approach for training language models.

- Evaluate on **unseen tasks**

Prompt *Explain the moon landing to a 6 year old in a few sentences.*

Completion GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

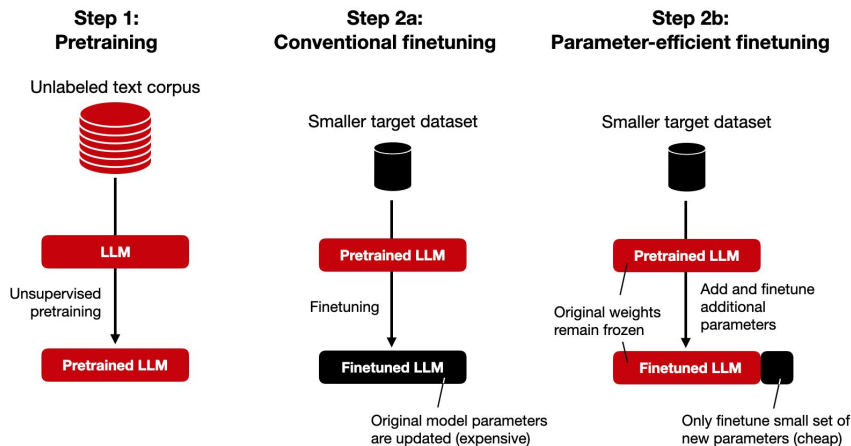
InstructGPT

People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.

<https://openai.com/research/instruction-following>

More efficient finetuning approaches: PEFT

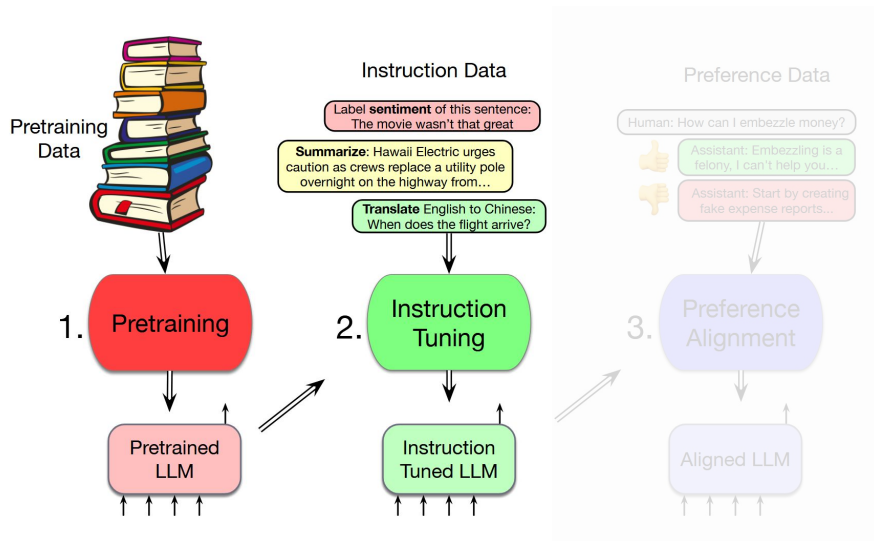
- As models get larger and larger, full fine-tuning becomes infeasible
- Instead, add a small number of new parameters to a pretrained LLM
- Only finetune the newly added parameters → **Parameter Efficient Finetuning (PEFT)**
- Common approaches: Low-Rank Adaptation (LoRA),



<https://magazine.sebastianraschka.com/p/finetuning-llms-with-adapters>

We now have all the ingredients for step 2

- A pretrained model that knows many aspects of language
- That has now been fine-tuned to follow and answer to instructions
- How do we get this model to ‘behave’?



Jurafsky and Martin, 2025

Reinforcement Learning from Human Feedback (RLHF)

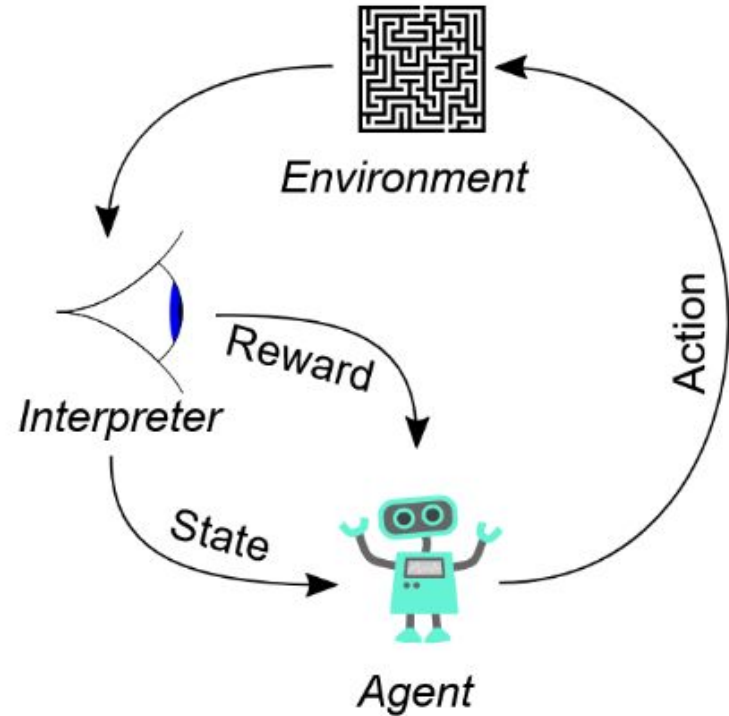
- Instruction finetuning is great but has some limitations
- Even with instruction finetuning, there a mismatch between the LM objective and the objective of “satisfy human preferences!”
- Can we explicitly attempt to satisfy human preferences?

Reinforcement Learning from Human Feedback (RLHF)

- Instruction finetuning is great but has some limitations
- Even with instruction finetuning, there a mismatch between the LM objective and the objective of “satisfy human preferences!”
- Can we explicitly attempt to satisfy human preferences?
- **Reinforcement Learning:** area of machine learning and optimal control concerned with how an intelligent agent ought to take actions in a dynamic environment in order to maximize the cumulative reward.

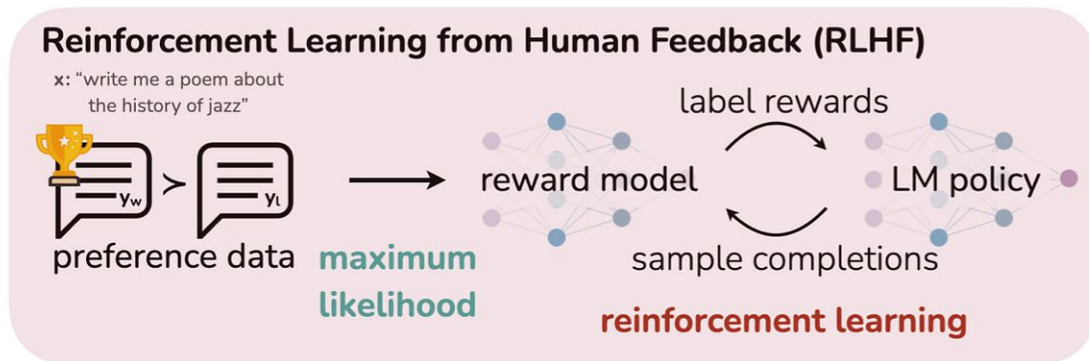
Detour: Reinforcement Learning

- an agent takes actions in an environment, which is interpreted into a reward and a state representation, which are fed back to the agent.
- **Reward model** that steers the model



How does RLHF work for LLMs?

- Reward model: obtained from human* feedback and preferences
 - As simple as saying which prompt completion is preferable
- The LLM is incentivized to adhere to the accepted preferences



*which humans?

Preference Datasets

Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback

Yuntao Bai*, Andy Jones, Kamal Ndousse,

Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort,
Deep Ganguli, Tom Henighan, Nicholas Joseph, Saurav Kadavath, Jackson Kernion,

Tom Conerly, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds,
Danny Hernandez, Tristan Hume, Scott Johnston, Shauna Kravec, Liane Lovitt,
Neel Nanda, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark,
Sam McCandlish, Chris Olah, Ben Mann, Jared Kaplan*

Anthropic

Abstract

We apply preference modeling and reinforcement learning from human feedback (RLHF) to finetune language models to act as helpful and harmless assistants. We find this alignment training improves performance on almost all NLP evaluations, and is fully compatible with training for specialized skills such as python coding and summarization. We explore an iterated online mode of training, where preference models and RL policies are updated on a weekly cadence with fresh human feedback data, efficiently improving our datasets and models. Finally, we investigate the robustness of RLHF training, and identify a roughly linear relation between the RL reward and the square root of the KL divergence between the policy and its initialization. Alongside our main results, we perform peripheral analyses on calibration, competing objectives, and the use of OOD detection, compare our models with human writers, and provide samples from our models using prompts appearing in recent related work.

Playground task ⓘ

Talk to the assistant

Task 1 of 5

Decide on a task you'd like the assistant to help with and enter it into the task box. Interact with the AI assistant. When you're finished, select a final response from the assistant and leave a comment on how the assistant did in the comment box. Click to go to the next task.

Comments

I thought the assistant was ...

Rating

☐ Bad ☐ ☐ ☐ ☐ Good

Next Task

Human

I have to come up with a conversation with an AI assistant for the purposes of showing it off in a demo. What sorts of conversation topics would you recommend?

Assistant

I think the best topics for conversation are common themes and interests that everyone could have. Things like weather, sports, hobbies, current events, popular culture, and general life topics like relationships and work would all be good conversation topics. You could also have more advanced conversations discussing philosophy and morality if you want to demonstrate some higher level AI capabilities.

Human

I think I'm more interested in an advanced conversation. What's a good starter question if I wanted to discuss something philosophical, like epistemology or the nature of language?

Assistant

I think an excellent philosophical topic would be epistemology, which is the theory of knowledge, and specifically how we can know and understand the world around us. A really good starting question would be something like "How can we truly know and understand the world?", or "What is the nature of knowledge?". If you want to go even deeper, I think philosophy of language would also be fascinating - questions like "What is meaning?", "How do language and thought relate?", and "How can language mislead and deceive?" would be great discussion topics.

Human

How would you answer a question like: How do language and thought relate?

Choose the most helpful and honest response

A

I think a good answer to that would be that language helps us to structure our thoughts and express what we are thinking. Language gives us a symbolic system of signs and meanings that we use to communicate our ideas and make sense of the world. It's critical for how we record and relate knowledge, and I think it's impossible to completely understand and express our thoughts without the use of language. Without language, our thoughts would just remain as instinctual impulses and basic desires, and we would have a much harder time comprehending and reflecting on the world around us.

B

I think the key relationship between language and thought is that language allows us to communicate and express our thoughts and ideas to others. Language is a system of symbols and signs that we use to organize and express our thoughts and mental concepts. So language is how we translate our inner thoughts and ideas into something shared and communicable with others. Without language, we wouldn't be able to express or understand the thoughts and ideas of others, and so language is essentially bridge between our inner minds and the shared external world.

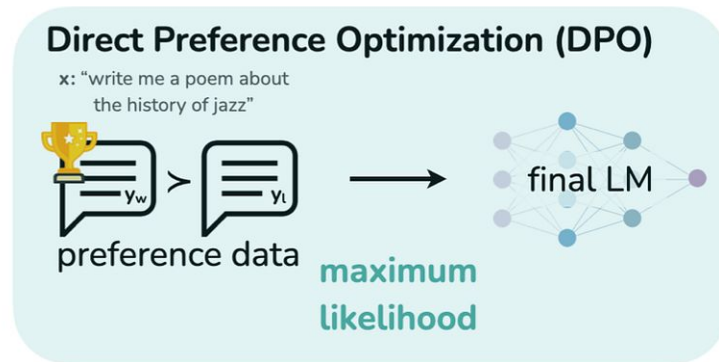
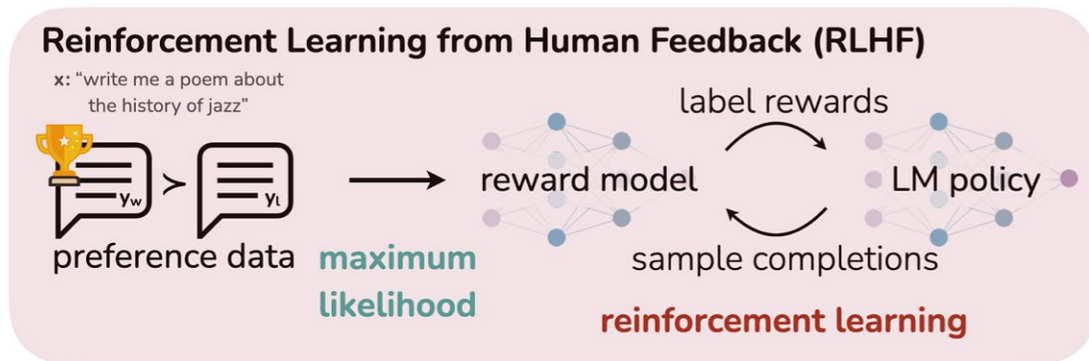
A A A A B B B B

A is better B is better

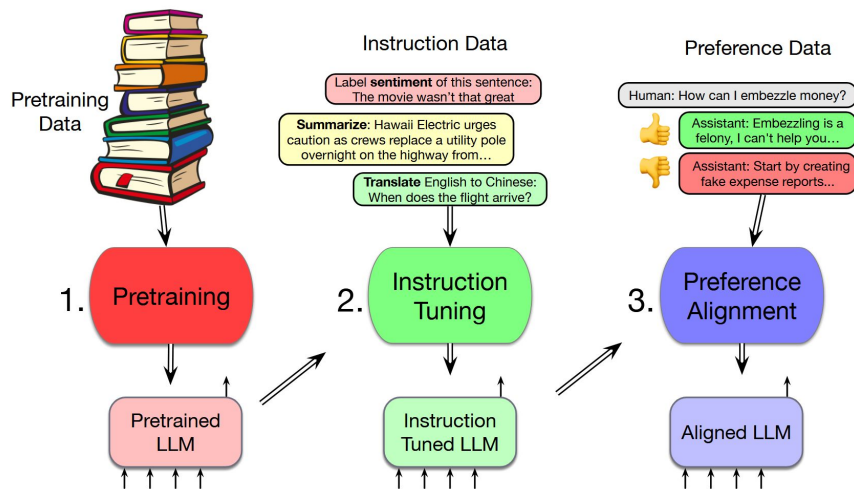
Figure 6 We show the interface that crowdworkers use to interact with our models. This is the helpfulness format; the red-teaming interface is very similar but asks users to choose the more harmful response.

RLHF is no longer main technique used in preference alignment

- The RL loop is both expensive and unnecessary
- We can directly optimize on preferences — Direct Policy Optimization (DPO, [Rafialov et al., 2023](#))
- Other options: **RL** with Verifiable Rewards (RLVR)

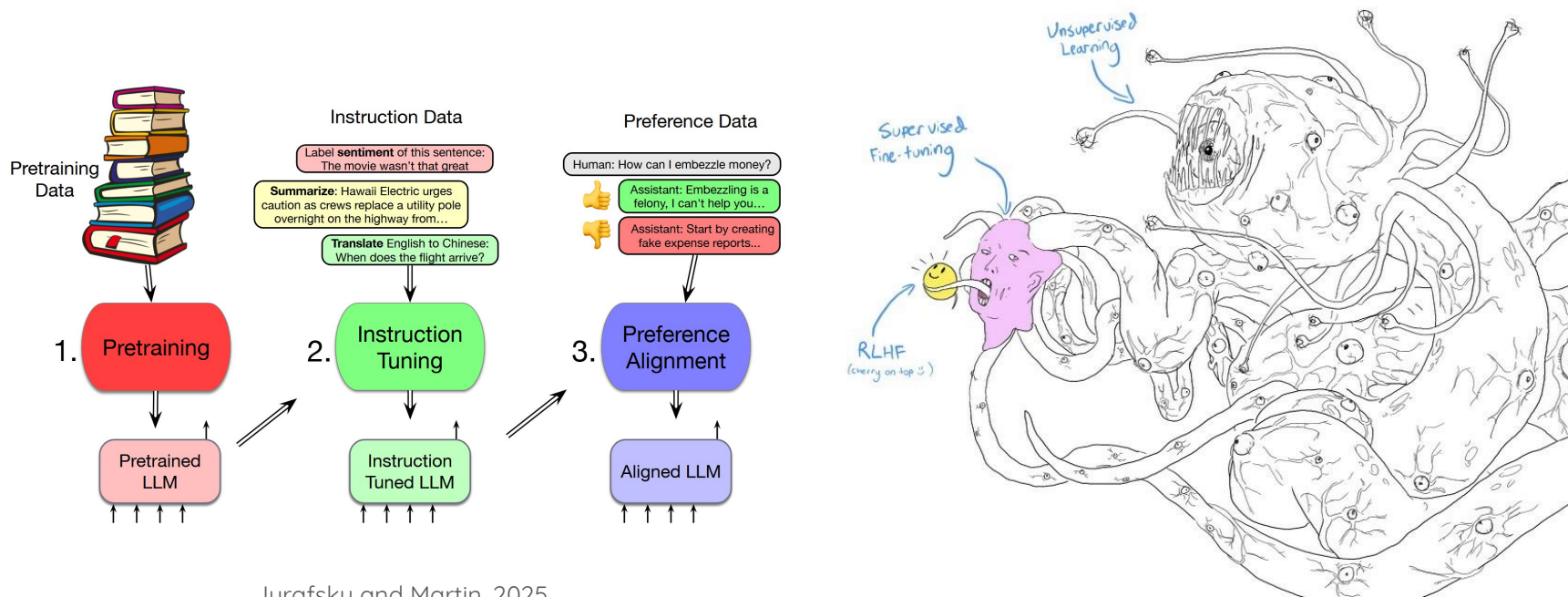


And so we have Large Language Models which we can prompt



Jurafsky and Martin, 2025

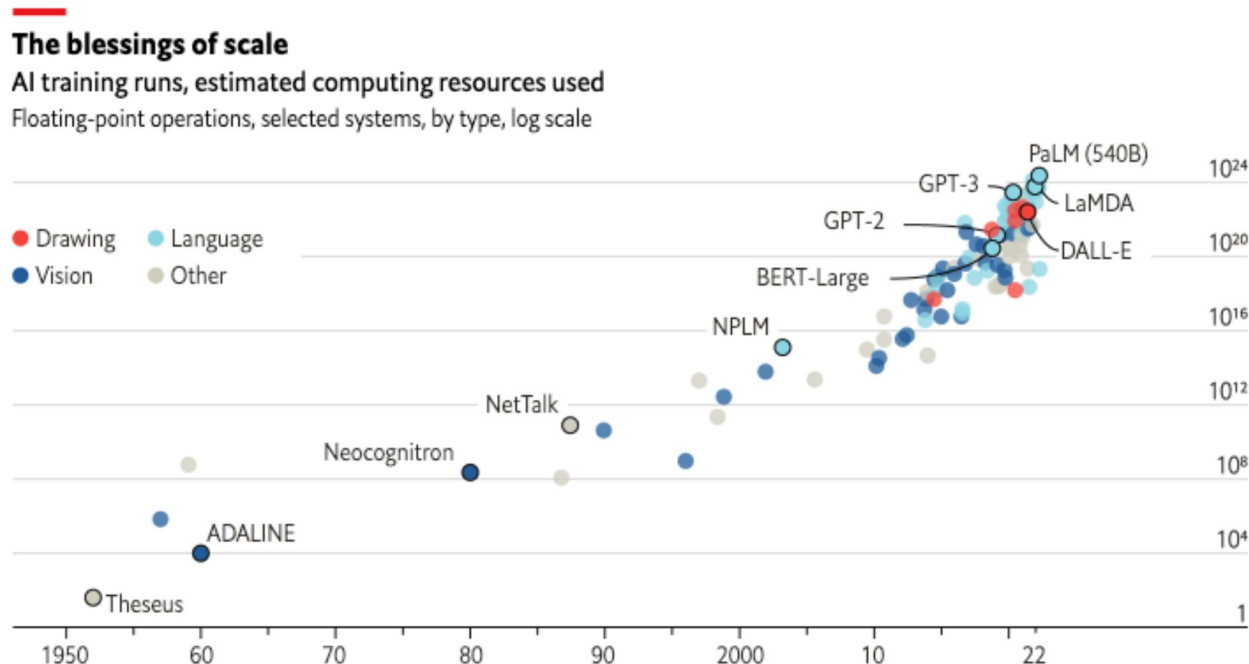
And so we have Large Language Models which we can prompt



Jurafsky and Martin, 2025

Can LLMs learn to do things it has never
been trained on?

Emergent Properties of LMs



Sources: "Compute trends across three eras of machine learning", by J. Sevilla et al., arXiv, 2022; Our World in Data

Emergent Properties of LMs

One key emergent ability in GPT-2 is **zero-shot** learning:

the ability to do many tasks with no examples, and no gradient updates, by simply:

- Specifying the right sequence prediction problem (e.g. question answering):

Passage: Tom Brady... Q: Where was Tom Brady born? A: ...

[Language models are unsupervised multitask learners](#)

Language Models are Unsupervised Multitask Learners

Alec Radford^{*1} Jeffrey Wu^{*1} Rewon Child¹ David Luan¹ Dario Amodei^{**1} Ilya Sutskever¹

Abstract

Natural language processing tasks, such as question answering, machine translation, reading comprehension, and summarization, are typically approached with supervised learning on task-specific datasets. We demonstrate that language models begin to learn these tasks without any explicit supervision when trained on a new dataset of millions of webpages called WebText. When conditioned on a document plus questions, the answers generated by the language model reach 55 F1 on the CoQA dataset - matching or exceeding the performance of 3 out of 4 baseline systems without using the 127,000+ training examples. The capacity of the language model is essential to the success of zero-shot task transfer and increasing it improves performance in a log-linear fashion across tasks. Our largest model, GPT-2,

competent generalists. We would like to make general systems which can perform many tasks without the need to manually create and train a dataset for each one.

The dominant approach to creating ML systems is to select a dataset of training examples demonstrating the desired behavior for a desired task, train a system to learn these behaviors, and then test its performance on a new dataset of examples. This approach has served well to make progress on many tasks, but the often erratic behavior of captioning models (Ji et al., 2017), reading comprehension systems (Jawahar et al., 2017), and image classifiers (Alcorn et al., 2018) on a wide variety of possible inputs highlights the limitations of this approach.

Our suspicion is that the prevalence of such behavior on single domain datasets is a major contributor to the lack of generalization observed in current systems.

Emergent zero-shot and few-shot learning

Specify a task by simply prepending examples of the task before your example

- Also called **in-context learning**, to stress that no parameter updates are performed when learning a new task

Language Models are Few-Shot Learners

Tom B. Brown*	Benjamin Mann*	Nick Ryder*	Melanie Subbiah*
Jared Kaplan†	Prafulla Dhariwal	Arvind Neelakantan	Pranav Shyam
Girish Sastry	Amanda Askell	Sandhini Agarwal	Ariel Herbert-Voss
Gretchen Krueger	Tom Henighan	Rewon Child	Aditya Ramesh
Daniel M. Ziegler	Jeffrey Wu	Clemens Winter	
Christopher Hesse	Mark Chen	Eric Sigler	Mateusz Litwin
Scott Gray			
Benjamin Chess	Jack Clark	Christopher Berner	
Sam McCandlish	Alec Radford	Ilya Sutskever	Dario Amodei

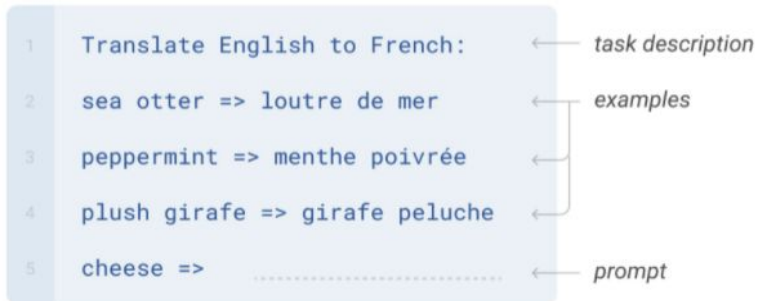
Abstract

We demonstrate that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even becoming competitive with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question-answering, and cloze tasks. We also identify some datasets where GPT-3's few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora.

'In-Context' Learning

- The model learns without any parameter updates => the underlying model is not retrained
- Prompt engineering: the process of finding effective prompts for a task.

Zero/few-shot prompting



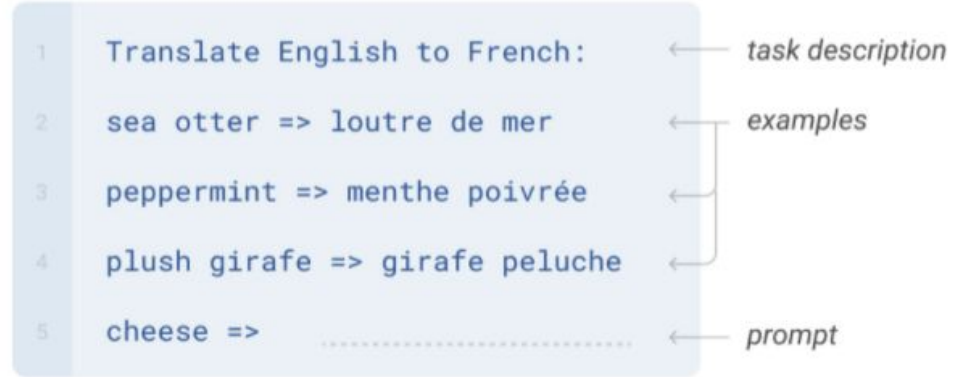
Traditional fine-tuning



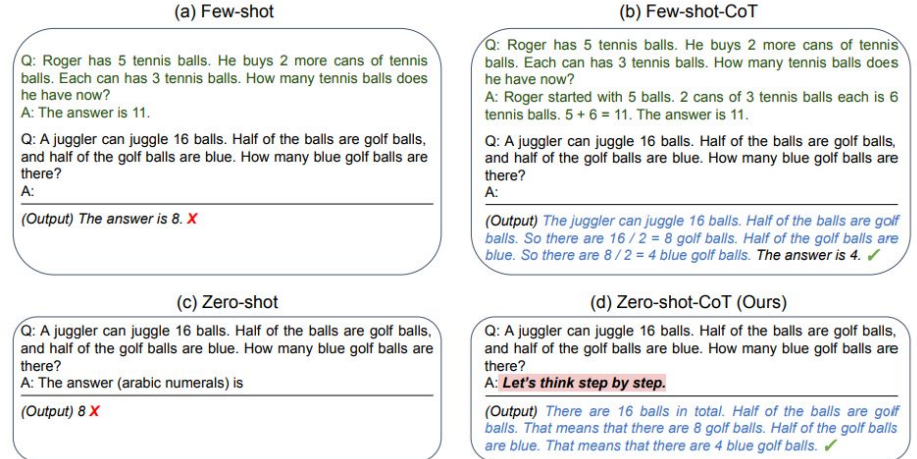
<https://web.stanford.edu/class/cs224n/slides/cs224n-2023-lecture11-prompting-rlhf.pdf>

Prompt based labeling

- **Prompt:** a text string that a user issues to a language model to get the model to do something useful by conditional generation
- Zero-shot: only instructions, no examples
- Few-shot: instructions + (a few) examples
- More advanced prompting strategies (more in the next lecture):
 - Chain-of-thought reasoning (COT)
 - Self-Consistency



<https://huggingface.co/blog/few-shot-learning-gpt-neo-and-inference-api>



<https://www.promptingguide.ai/techniques/cot>

Group discussion: Think-Pair-Share

Based on what you saw so far and last week's reading:

- a. In your experience, what are tasks LLMs excel at and those they struggle with?
 - i. Do you have a hypothesis of **why** this might be?
 - ii. Try to see if you can relate the strengths and weaknesses of LLMs to how they are built.
- b. What is the intuition behind the **attention** mechanism? How does it obtain meaning from tokens?

Summary and Next Week

- This session we learned:
 - How LLMs generate content
 - How LLMs follow instructions
 - How LLMs learn to ‘behave’
- Next time:
 - All about prompting LLMs
 - Project information
 - Please read: [The Intuition Behind How Large Language Models Work, Part II](#)

Questions?

Additional Resources

- AI for Humanists: BERT workshop [<https://aiforhumanists.com/tutorials/>]
- Text Generation Primer: <https://huggingface.co/tasks/text-generation>
- RLHF Primer: <https://huggingface.co/blog/rlhf>
- Daniel Jurafsky and James H. Martin. 2025. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models, <https://web.stanford.edu/~jurafsky/slp3>
 - <https://web.stanford.edu/~jurafsky/slp3/slides/llm25aug.pdf>
 - A more detailed primer of posttraining: <https://web.stanford.edu/~jurafsky/slp3/8.pdf>